



Guangdong Technion

Israel Institute of Technology

广东以色列理工学院



Tutorial Course for Intro to Sensor-Integrated Systems

Qilong Zhong, MSc

Mechanical Engineering and Robotics

Guangdong Technion - Israel Institute of Technology

qilong.zhong@gtiit.edu.cn

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Exercise

A force sensor has a mass of 0.5 kg, stiffness of $2 \times 10^2 \text{ N m}^{-1}$ and a damping constant of 6.0 N s m^{-1} .

- (a) Calculate the steady-state sensitivity, natural frequency and damping ratio for the sensor.
- (b) Calculate the displacement of the sensor for a steady input force of 2 N.

Solution

For a second-order mechanical system (mass-spring-damper system), the governing differential equation is:

$$m \frac{d^2 x}{dt^2} + c \frac{dx}{dt} + kx = F$$

(b) Calculate the displacement for a steady input force of 2 N

$$x = K \cdot F = 0.005 \times 2 = 0.01 \text{ m}$$

1. Steady-state Sensitivity (K)

$$K = \frac{x}{F} = \frac{1}{k}$$

$$K = \frac{1}{200} = 0.005 \text{ m N}^{-1}$$

2. Natural Frequency (ω_n)

The undamped natural frequency is determined by the stiffness and the mass of the system:

$$\omega_n = \sqrt{\frac{k}{m}}$$

$$\omega_n = \sqrt{\frac{200}{0.5}} = \sqrt{400} = 20 \text{ rad s}^{-1}$$

3. Damping Ratio (ξ)

The damping ratio relates the actual damping to the critical damping of the system, given by the formula:

$$\xi = \frac{c}{2\sqrt{mk}} \quad \xi = \frac{6.0}{2\sqrt{0.5 \times 200}} = \frac{6.0}{2\sqrt{100}} = \frac{6.0}{2 \times 10} = 0.3$$

Exercise

An elastic force sensor has an effective seismic mass of 0.1 kg, a spring stiffness of 10 N m^{-1} and a damping constant of 14 N s m^{-1} .

- (a) Calculate the following quantities:
- (i) sensor natural frequency
 - (ii) sensor damping ratio
 - (iii) transfer function relating displacement and force.

Solution

(a) Sensor Natural Frequency (ω_n)

The undamped natural frequency is determined by the stiffness and the mass of the system:

$$\omega_n = \sqrt{\frac{k}{m}}$$

Substituting the given values:

$$\omega_n = \sqrt{\frac{10}{0.1}} = \sqrt{100} = 10 \text{ rad/s}$$

(b) Sensor Damping Ratio (ξ)

The damping ratio relates the actual damping to the critical damping of the system, given by the formula:

$$\xi = \frac{c}{2\sqrt{mk}}$$

Substituting the given values:

$$\xi = \frac{14}{2\sqrt{0.1 \times 10}} = \frac{14}{2\sqrt{1}} = \frac{14}{2} = 7$$

(c) Transfer Function relating input force to output displacement

The governing differential equation for a standard second-order mass-spring-damper system is:

Substituting the numerical values ($m = 0.1$, $c = 14$, $k = 10$):

$$G(s) = \frac{1}{0.1s^2 + 14s + 10} \quad G(s) = \frac{10}{s^2 + 140s + 100}$$

Exercise

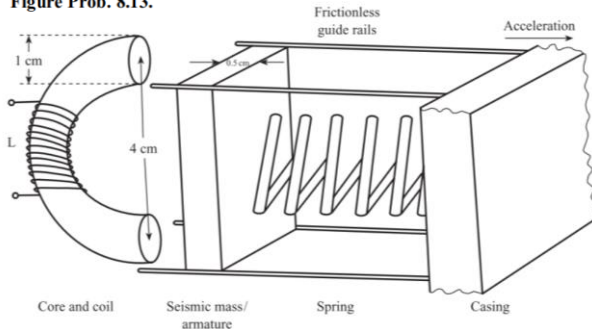
A seismic mass accelerometer with a variable reluctance displacement sensor is shown in Figure Prob. 13. The undamped natural frequency of the accelerometer is to be 25 Hz and the range of the acceleration input is to be 0 to 5g. Using the data given below, calculate:

- (a) the required spring stiffness;
- (b) the air gap at maximum acceleration;
- (c) the range of variation of the coil inductance.

Data

Relative permeability of armature and coil material	= 100
Relative permeability of air	= 1
Permeability of free space	= $4\pi \times 10^{-7}$ H/m
Effective mass of armature and spring	= 0.025 kg
Acceleration of gravity g	= 9.81 m/s^2
Number of coil turns	= 500
Air gap at zero acceleration	= 3 mm

Figure Prob. 8.13.



Solution

(a) Calculate the required spring stiffness (k)

The relationship between the undamped natural frequency f_n , the spring stiffness k , and the mass m of the accelerometer is given by:

$$f_n = \frac{1}{2\pi} \sqrt{\frac{k}{m}}$$

First, calculate the angular frequency ω_n :

$$\omega_n = 2\pi f_n = 2\pi \times 25 = 50\pi \text{ rad/s}$$

Then, rearrange the formula to solve for k :

$$k = m\omega_n^2 \quad k \approx 616.85 \text{ N/m}$$

(b) Calculate the air gap at maximum acceleration

According to the diagram, when the casing accelerates to the right, the inertial force will cause the seismic mass (armature) to move to the left relative to the casing, thereby **decreasing** the air gap.

Maximum acceleration $a_{max} = 5g$:

$$a_{max} = 5 \times 9.81 = 49.05 \text{ m/s}^2$$

The maximum inertial force F_{max} is:

$$F_{max} = m \cdot a_{max} = 0.025 \times 49.05 = 1.22625 \text{ N}$$

The maximum displacement of the mass x_{max} is:

$$x_{max} = \frac{F_{max}}{k} = \frac{1.22625}{616.85} \approx 0.001988 \text{ m} = 1.988 \text{ mm}$$

The air gap at maximum acceleration d_{min} is the initial air gap minus the displacement:

$$d_{min} = d_0 - x_{max} = 3 \text{ mm} - 1.988 \text{ mm} = 1.012 \text{ mm}$$

Solution

(c) Calculate the range of variation of the coil inductance

To calculate the inductance L , we need to analyze the total reluctance R_{total} of the magnetic circuit. The formula for inductance is:

$$L = \frac{N^2}{R_{total}}$$

The total reluctance consists of the core reluctance R_c , the armature reluctance R_a , and the reluctance of the two air gaps R_g : $R_{total} = R_c + R_a + R_g$.

○ Core Reluctance R_c :

$$R_c = \frac{l_c}{\mu_0 \mu_r A_c} = \frac{0.02\pi}{\mu_0 \times 100 \times \left(\frac{\pi \times 10^{-4}}{4}\right)} = \frac{8}{\mu_0}$$

○ Armature Reluctance R_a :

$$R_a = \frac{l_a}{\mu_0 \mu_r A_a} = \frac{0.04}{\mu_0 \times 100 \times (5 \times 10^{-5})} = \frac{8}{\mu_0}$$

○ Total Reluctance of the Iron Parts R_{iron} :

$$R_{iron} = R_c + R_a = \frac{16}{\mu_0}$$

○ Air Gap Reluctance R_g (two air gaps, each with length d):

$$R_g = \frac{2d}{\mu_0 A_c} = \frac{2d}{\mu_0 \times \left(\frac{\pi \times 10^{-4}}{4}\right)} = \frac{80000d}{\pi \mu_0}$$

3. Total Reluctance and Inductance Formulas:

$$R_{total} = \frac{1}{\mu_0} \left(16 + \frac{80000d}{\pi} \right)$$

$$L(d) = \frac{N^2 \mu_0}{16 + \frac{80000d}{\pi}} = \frac{500^2 \times 4\pi \times 10^{-7}}{16 + \frac{80000d}{\pi}} = \frac{0.1\pi}{16 + \frac{80000d}{\pi}}$$

4. Calculate the Range of Inductance Variation:

○ At 0 acceleration ($d_0 = 0.003$ m):

$$L_{min} = \frac{0.1\pi}{16 + \frac{80000 \times 0.003}{\pi}} = \frac{0.1\pi}{16 + \frac{240}{\pi}} \approx \frac{0.314159}{16 + 76.394} \approx 0.00340 \text{ H} = 3.40 \text{ mH}$$

○ At maximum acceleration of 5g ($d_{min} = 0.001012$ m):

$$L_{max} = \frac{0.1\pi}{16 + \frac{80000 \times 0.001012}{\pi}} = \frac{0.1\pi}{16 + \frac{80.96}{\pi}} \approx \frac{0.314159}{16 + 25.770} \approx 0.00752 \text{ H} = 7.52 \text{ mH}$$

The range of variation of the coil inductance is 3.40 mH to 7.52 mH.

Exercise

A steel bar has a cross-sectional area of 10^{-3} m^2 , Young's modulus of $2 \times 10^{11} \text{ Pa}$ and Poisson's ratio of 0.4. If the bar is subject to a compressive force of 10^5 N , find the corresponding longitudinal and transverse strains.

Solution

Step 1: Calculate Longitudinal Stress (σ)

Stress is equal to the applied force divided by the cross-sectional area:

$$\sigma = \frac{F}{A}$$

$$\sigma = \frac{-10^5 \text{ N}}{10^{-3} \text{ m}^2} = -10^8 \text{ Pa}$$

Step 2: Calculate Longitudinal Strain (ε_L)

According to Hooke's Law, within the elastic limit, stress is proportional to strain, and the constant of proportionality is Young's modulus:

$$\varepsilon_L = \frac{\sigma}{E}$$

$$\varepsilon_L = \frac{-10^8 \text{ Pa}}{2 \times 10^{11} \text{ Pa}} = -0.5 \times 10^{-3} = -5 \times 10^{-4}$$

Step 3: Calculate Transverse Strain (ε_T)

Poisson's ratio describes the ratio of transverse deformation to longitudinal deformation when a material is subjected to uniaxial stress. Its formula is:

$$\nu = -\frac{\varepsilon_T}{\varepsilon_L}$$

By rearranging the formula, we can solve for the transverse strain:

$$\varepsilon_T = -\nu \cdot \varepsilon_L$$

$$\varepsilon_T = -0.4 \times (-5 \times 10^{-4}) = 2 \times 10^{-4}$$

Exercise

Two strain gauges, each with an unstrained resistance of $120\ \Omega$ and a gauge factor of 2.0, are bonded onto the steel bar of Problem 8.17 so that one measures the longitudinal strain and one the transverse strain. Find the corresponding gauge resistances for the longitudinal and transverse strains.

Solution

The change in resistance of a strain gauge (ΔR) is related to the initial resistance, gauge factor, and strain by the formula:

$$\Delta R = R_0 \cdot G \cdot \varepsilon$$

The total resistance (R) of the strain gauge after deformation is:

$$R = R_0 + \Delta R$$

1. Calculate the resistance of the longitudinal strain gauge (R_L)

First, calculate the change in resistance caused by the longitudinal strain:

$$\Delta R_L = 120 \times 2.0 \times (-5 \times 10^{-4})$$

$$\Delta R_L = 240 \times (-0.0005) = -0.12 \, \Omega$$

Calculate the total resistance of the longitudinal strain gauge at this point:

$$R_L = R_0 + \Delta R_L = 120 - 0.12 = 119.88 \, \Omega$$

2. Calculate the resistance of the transverse strain gauge (R_T)

First, calculate the change in resistance caused by the transverse strain:

$$\Delta R_T = 120 \times 2.0 \times (2 \times 10^{-4})$$

$$\Delta R_T = 240 \times 0.0002 = 0.048 \, \Omega$$

Calculate the total resistance of the transverse strain gauge at this point:

$$R_T = R_0 + \Delta R_T = 120 + 0.048 = 120.048 \, \Omega$$

Exercise

Four strain gauges, with specification given below, are available to measure the torque on a cylindrical shaft 4 cm in diameter connecting a motor and load.

- (a) Draw clearly labelled diagrams showing:
- (i) the arrangement of the gauges on the shaft;
 - (ii) the arrangement of the gauges in the bridge circuit, for optimum accuracy and sensitivity.
- (b) Calculate the maximum achievable bridge out-of-balance voltage for an applied torque T of 10^3 N m given the following:

Tensile and compressive strains $= \pm T/\pi S a^3$ where $S = 1.1 \times 10^{11}$ N m⁻² is the shear modulus of the shaft material and a is the radius of the shaft in metres.

Strain gauge data:

Resistance = 120 Ω

Gauge factor = 2.1

Maximum current = 50 mA

Solution

Step 1: Calculate the principal strain (ϵ)

Using the formula provided in the problem:

$$\epsilon = \frac{T}{\pi S a^3}$$

Substitute the values:

$$\epsilon = \frac{10^3}{\pi \times 1.1 \times 10^{11} \times (0.02)^3}$$
$$\epsilon = \frac{1000}{8.8\pi \times 10^5} \approx 3.617 \times 10^{-4}$$

Step 2: Calculate the maximum allowable bridge excitation voltage (V_s)

In a Wheatstone bridge, the power supply is connected across two parallel branches. Each branch has two strain gauges in series.

Total resistance of one branch = $R + R = 120 \, \Omega + 120 \, \Omega = 240 \, \Omega$.

To ensure the current through the gauges does not exceed 50 mA (0.05 A), the maximum supply voltage V_s is:

$$V_s = I_{max} \times (\text{Branch Resistance}) = 0.05 \, \text{A} \times 240 \, \Omega = 12 \, \text{V}$$

Step 3: Calculate the maximum out-of-balance voltage (V_{out})

$$V_{out} = V_s \cdot G \cdot \epsilon$$

Substitute our calculated values:

$$V_{out} = 12 \times 2.1 \times (3.617 \times 10^{-4})$$
$$V_{out} \approx 25.2 \times 3.617 \times 10^{-4} \approx 0.009115 \, \text{V}$$

Exercise

A load cell consists of a domed vertical steel cylinder 20 cm high and 15 cm in diameter. Four flat surfaces, at right angles to each other, are cut on the vertical surface so as to form 10 cm squares. Resistance strain gauges are attached to these flat surfaces so that two gauges (on opposite faces) suffer longitudinal compression and two gauges (on the other pair of opposite faces) suffer transverse tension. The strain gauges have the following specification:

Resistance	= 100 Ω
Gauge factor	= 2.1
Maximum gauge current	= 30 mA

The gauges are connected in a temperature-compensated bridge and the out-of-balance signal is input to a differential amplifier. Calculate the minimum amplifier gain if the amplifier output voltage is to be 1 V for a compressive force of 10^5 N.

Young's modulus for steel	= 2.1×10^{11} N m ⁻²
Poisson's ratio for steel	= 0.29

Solution

Step 1: Calculate the effective cross-sectional area (A)

Based on the problem description, the load-bearing cross-section is a $10\text{ cm} \times 10\text{ cm}$ square.

$$A = 0.1\text{ m} \times 0.1\text{ m} = 0.01\text{ m}^2$$

Step 2: Calculate longitudinal strain (ϵ_L) and transverse strain (ϵ_T)

First, calculate the compressive stress σ :

$$\sigma = \frac{F}{A} = \frac{10^5}{0.01} = 10^7\text{ N/m}^2$$

According to Hooke's Law, calculate the longitudinal strain (compression, taken as an absolute value for the formula):

$$\epsilon_L = \frac{\sigma}{E} = \frac{10^7}{2.1 \times 10^{11}} = \frac{1}{21000} \approx 4.7619 \times 10^{-5}$$

According to Poisson's ratio, calculate the transverse strain (tension, opposite sign to longitudinal):

$$\epsilon_T = \nu \cdot \epsilon_L = 0.29 \times \frac{1}{21000} \approx 1.3810 \times 10^{-5}$$

Step 3: Calculate the maximum bridge excitation voltage (V_s)

A full Wheatstone bridge circuit consists of two parallel branches. Each branch contains two strain gauges in series, so the total resistance of a single branch is $2R = 200\ \Omega$.

To ensure the current does not exceed the maximum allowable gauge current (0.03 A), the maximum supply voltage across the bridge is:

$$V_s = I_{max} \times (2R) = 0.03 \times 200 = 6\text{ V}$$

Step 4: Calculate the bridge out-of-balance voltage (V_{out})

$$V_{out} = \frac{V_s}{4} \cdot G \cdot [\epsilon_1 - \epsilon_2 + \epsilon_3 - \epsilon_4]$$

Solution

Substituting the two $+\epsilon_T$ and two $-\epsilon_L$ strains (note that they subtract algebraically because they are on adjacent arms):

$$V_{out} = \frac{V_s}{4} \cdot G \cdot [\epsilon_T - (-\epsilon_L) + \epsilon_T - (-\epsilon_L)] = \frac{V_s}{2} \cdot G \cdot (\epsilon_L + \epsilon_T)$$

Substituting $\epsilon_T = \nu\epsilon_L$ gives:

$$V_{out} = \frac{V_s}{2} \cdot G \cdot \epsilon_L \cdot (1 + \nu)$$

Substitute the specific numerical values:

$$V_{out} = \frac{6}{2} \times 2.1 \times \frac{1}{21000} \times (1 + 0.29)$$
$$V_{out} = 3 \times 2.1 \times \frac{1.29}{21000} = \frac{8.127}{21000} \approx 3.87 \times 10^{-4} \text{ V}$$

Step 5: Calculate the required amplifier gain (K)

The differential amplifier needs to amplify the weak out-of-balance signal from the bridge to the required 1 V.

$$K = \frac{V_{amp_out}}{V_{out}} = \frac{1}{3.87 \times 10^{-4}}$$

$$K \approx 2583.98$$